Experiment No.: 03
Title: To design and implement Decision Tree classifiers

Batch: B3 **Roll No: 16014225801** **Experiment No: 02**

Aim: To implement a Decision Tree classifier for classifying data and evaluate its performance using appropriate evaluation metrics.

Resources needed: Python

Theory:

A Decision Tree is a supervised machine learning algorithm used for both classification and regression tasks. It predicts the value of a target variable by learning simple decision rules inferred from the training data. The model resembles a tree structure where each internal node represents a decision based on an attribute, each branch represents the outcome of the decision, and each leaf node represents the final class label.

Decision Trees are widely used because they are easy to understand, interpret, and visualize. The algorithm recursively partitions the dataset into subsets based on the feature that provides the highest information gain or lowest impurity.

The most commonly used Decision Tree algorithms include:

- CART (Classification and Regression Trees)
- C4.5
- ID3 (Iterative Dichotomiser 3)

Procedure:

1. Load and preprocess the dataset.
2. Select the best attribute as the root node.
3. Split the dataset based on the selected attribute.
4. Repeat the process recursively for each subset.
5. Stop splitting when all records belong to the same class or other stopping criteria are met.
6. Predict the class label for unseen data.

Results:

```
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree

df = pd.read_csv("response_1786094863508.csv")


X = df[["JAN-FEB", "MAR-MAY", "JUN-SEP", "OCT-DEC"]]
y = (df["ANNUAL"] >= df["ANNUAL"].median()).astype(int)


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

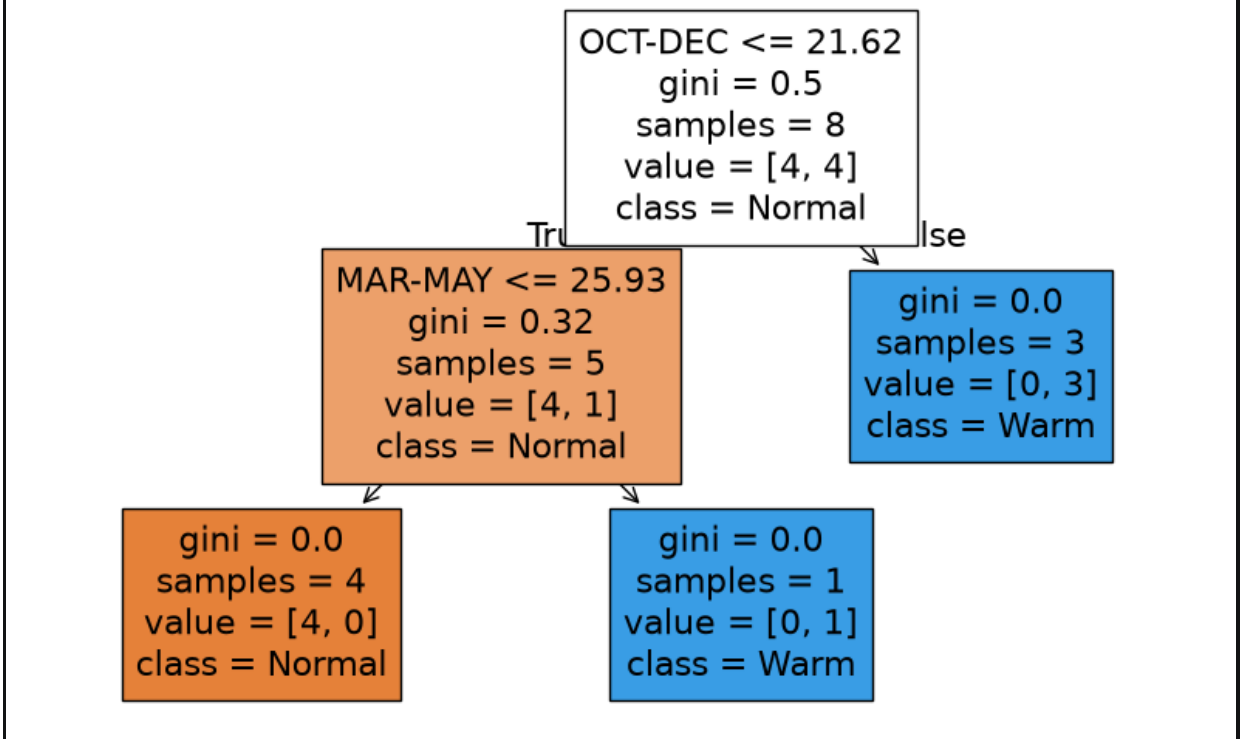
model = DecisionTreeClassifier(max_depth=2, random_state=42)
model.fit(X_train, y_train)
```

```
print("Accuracy:", model.score(X_test, y_test))

plt.figure(figsize=(10, 6))
plot_tree(
    model, feature_names=X.columns, class_names=["Normal", "Warm"], filled=True
)
plt.show()
```

Output:

```
Accuracy: 0.5
```



Questions & Answers:

1. What are the different Decision Tree algorithms?

Different Decision Tree algorithms include ID3, C4.5, C5.0, CART, and CHAID. ID3 uses Information Gain to select the best feature for splitting. C4.5 improves ID3 by using Gain Ratio and handling continuous values. C5.0 is a faster and more efficient version of C4.5. CART uses the Gini Index and creates binary splits, while CHAID uses statistical tests such as Chi-square to determine the best splits.

2. Why is feature selection important in Decision Trees?

Feature selection is important because it helps the Decision Tree choose the most useful attributes for making predictions. Selecting relevant features reduces unnecessary complexity, improves the model's accuracy and training speed, and makes the tree easier to understand. It can also reduce noise in the dataset and decrease the possibility of overfitting.

3. What causes overfitting in Decision Trees?

Overfitting occurs when a Decision Tree becomes too deep and complex and starts learning small details, noise, and unusual patterns from the training data. This usually happens when the tree

has too many branches, very small leaf nodes, or no restriction on its growth. As a result, the model performs very well on training data but performs poorly on new or unseen data.

4. How can Decision Tree overfitting be reduced?
Decision Tree overfitting can be reduced by controlling the size and complexity of the tree. Techniques such as pruning unnecessary branches, limiting the maximum tree depth, setting a minimum number of samples for splitting, and setting a minimum number of samples in leaf nodes can be used. Proper feature selection and sufficient training data also help the tree generalize better to unseen data.

Conclusion:
The Decision Tree classifier was successfully implemented for data classification, achieving measurable accuracy while demonstrating feature-based decision making and model evaluation techniques.

References:
1. Stephen Marsland, “Machine learning An Algorithmic Perspective CRC”, Press, 2015
2. Christopher M. Bishop “Pattern Recognition and Machine Learning Springer Science, Business Media 2006
3. Simon Haykin, “Neural Networks and Learning Machines” Pearson Education, 2009